

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

BINOMIAL & BETA

SANJOY DASGUPTA, PROFESSOR

UC San Diego

COMPUTER SCIENCE & ENGINEERING
HALICIOĞLU DATA SCIENCE INSTITUTE



The Beta Distribution

For $\alpha, \beta > 0$, the $\text{beta}(\alpha, \beta)$ distribution on $[0, 1]$ has functional form

$$p(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

where $\Gamma(\cdot)$ is the gamma function.

Recall $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$. Useful identity: $\Gamma(z+1) = z\Gamma(z)$.

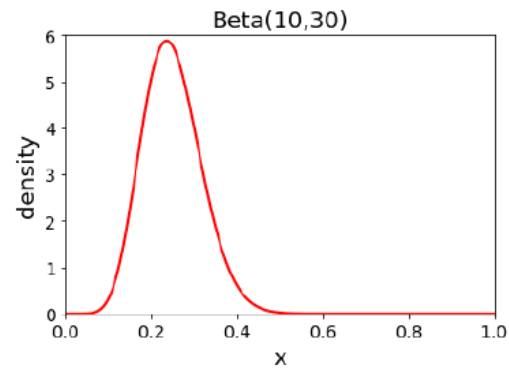
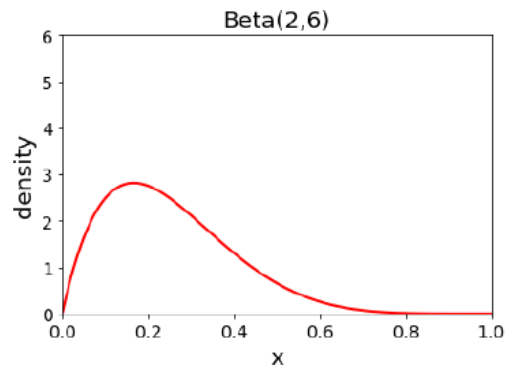
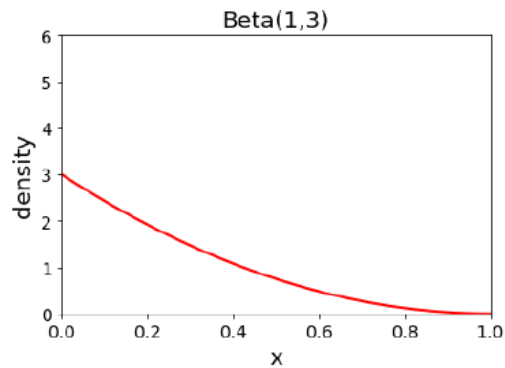
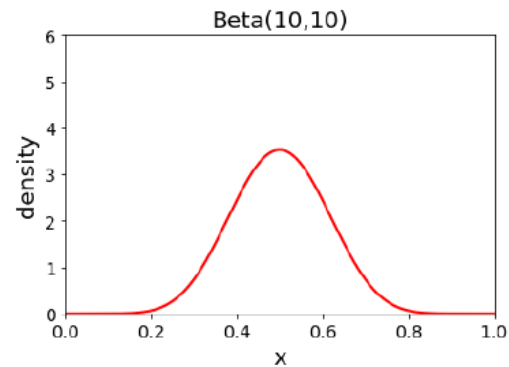
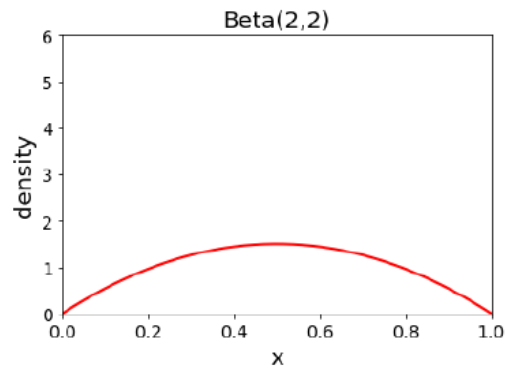
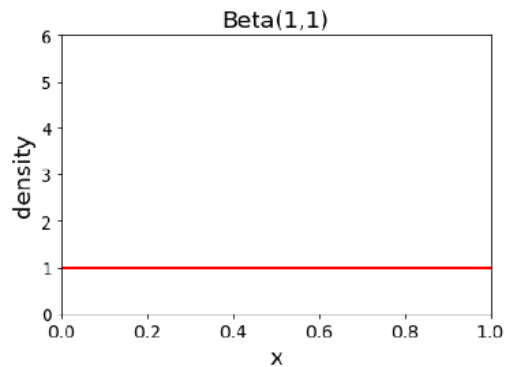
Basic properties of the $\text{beta}(\alpha, \beta)$ distribution:

$$\text{Mean} = \frac{\alpha}{\alpha + \beta}$$

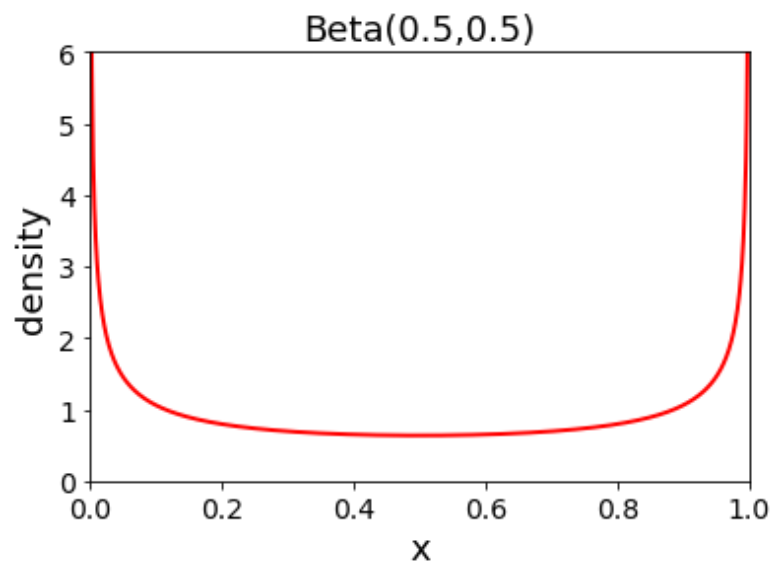
$$\text{Mode} = \frac{\alpha - 1}{\alpha + \beta - 2}$$

$$\text{Variance} = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

Beta Pictures



An Interesting Case



Laplace's Law of Succession

Another question that Laplace considered: What is the probability that the sun will rise tomorrow, given that it has risen every day for the past 5000 years?

- Let θ be the probability that the sun rises on any given day.
- Place a uniform prior on θ .
- We have $n = 5000 \times 365 = 1,826,213$ observations.
- Recall: **uniform prior + binomial likelihood $\Rightarrow$ beta posterior.**
- Posterior q_n is $\text{beta}(n + 1, 1)$.

Answer:

$$\Pr(\text{sun rises tomorrow}) = \int_0^1 \theta q_n(\theta) d\theta = \mathbb{E}_{q_n}[\theta] = \frac{n+1}{n+2}.$$

Beta Priors

Unknown probability $\theta \in [0,1]$ (of female child, sun rising, etc.)

- Prior $q_o = \text{beta}(\alpha, \beta)$
- See n observations, of which s are success
- What is the posterior q_n ?

The Beta is a Conjugate Prior for the Binomial

Conjugate prior: posterior is from the same family as the prior.

beta prior + binomial likelihood $\Rightarrow$ beta posterior

- Prior $\text{beta}(\alpha, \beta)$
- Observe s successes and f failures
- Posterior is $\text{beta}(\alpha + s, \beta + f)$

Equivalent sample size: prior $\text{beta}(\alpha, \beta)$ is the same as

- Uniform prior
- Seeing $\alpha - 1$ successes and $\beta - 1$ failures

Prior is eventually overwhelmed by observations.